Tutorial-01

1) A fitness app predicts whether a user will excercise outdoors based on weather conditions using a single layer Neural Network. Build network with 5 neurons in hidden layer.

x_1 : Temperature level (scaled)
x_2 : Humidity level (scaled)
x_3 : Bias (c.1)

Dataset

x_1	x_2	x_3	Target
7	2	1	1
3	5	1	0
6	3	1	1
2	6	1	0

Network Architecture

Input layer → 5 Hidden Neurons → 1 output layer

Activations Functions $\sigma(z) = \frac{1}{1+e^{-z}}$

Learning Parameters $\eta = 0.1$

Initial weights

h_1 $w_1 = 0.2$ $w_2 = 0.3$ $w_3 = 0.1$

h_2 $w_1 = 0.1$ $w_2 = 0.4$ $w_3 = 0.2$

h_3 $w_1 = 0.3$ $w_2 = 0.2$ $w_3 = 0.1$

h_4 $w_1 = 0.4$ $w_2 = 0.1$ $w_3 = 0.2$

h_5 $w_1 = 0.3$ $w_2 = 0.3$ $w_3 = 0.1$

output layer weights

$v_1 = 0.2,\ v_2 = 0.1,\ v_3 = 0.3,\ v_4 = 0.2,\ v_5 = 0.4,\ b = 0$

training example:-

$x_1 = 7,\ x_2 = 2,\ x_3 = 1$

$\Rightarrow h_1 = w_1 x_1 + w_2 x_2 + w_3 x_2$

$= 0.2 \times 7 + 0.3 \times 2 + 0.1 \times 5$

$z_1 = 2.1$ $\quad \sigma(z_1) = \frac{1}{1+e^{-z}} = 0.8909$

$\Rightarrow h_2 = w_4 x_1 + w_5 x_2 + w_6 x_3$

$= 0.1 \times 7 + 0.4 \times 2 + 0.2 \times 1$

$z = 1.7$ $\quad \sigma(z) = \frac{1}{1+e^{-1.7}} = 0.845$

$\Rightarrow h_3 = w_7 x_1 + w_8 x_2 + w_9 x_3$

$= 0.3 \times 7 + 0.2 \times 2 + 0.1 \times 2$

$z = 2.6$ $\quad \sigma(z) = \frac{1}{1+e^{-2.6}} = 0.930$

$\Rightarrow h_4 = w_{10} x_1 + w_{11} x_2 + w_{12} x_3$

$= 0.4 \times 7 + 0.1 \times 2 + 0.2 \times 1$

$z = 3.2$ $\quad \sigma(z) = \frac{1}{1+e^{-3.2}} = 0.960$

$\Rightarrow h_5 = w_{13} x_1 + w_{14} x_2 + w_{15} x_3$

$= 0.3 \times 7 + 0.3 \times 2 + 0.1 \times 1$

$z = 2.8$ $\quad \sigma(z) = \frac{1}{1+e^{-2.8}} = 0.9426$

$\rightarrow z_6 = 0.2(0.8909) + 0.1(0.845) + 0.3(0.936)$

$+ 0.2(0.960) + 0.4(0.94) + 0.1$

$z_6 = 1.20968$ $\quad \sigma(z) = \frac{1}{1+e^{1.20968}}$

$\sigma(z) = 0.77$